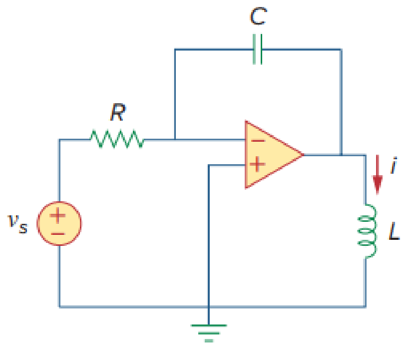
Problem

For the op amp circuit in Fig. 8.108, find the differential equation for $i(t)$.

Figure 8.108:



Step-by-step solution

Step 1 of 3

Consider the following circuit diagram.

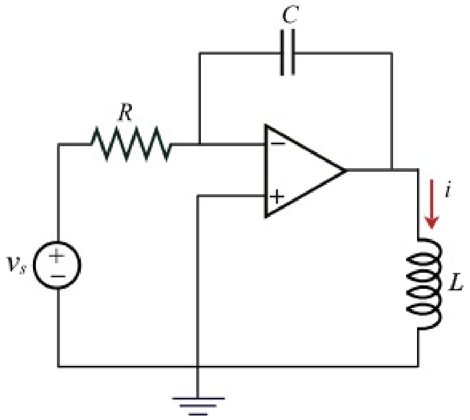


Figure 1

Step 2 of 3

The voltage at the inverting and the non-inverting terminals are equal and it is equal to zero in Figure 1 by the virtual ground concept.

Redraw the circuit in Figure 1 after indicating the voltages at the nodes.

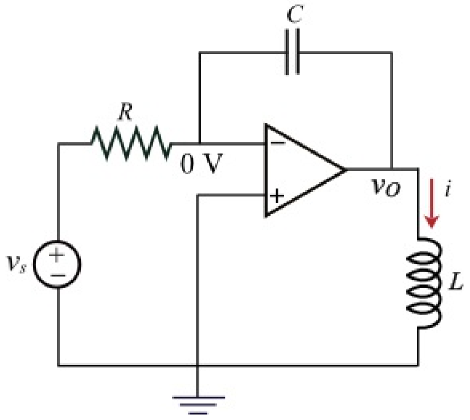


Figure 2

Step 3 of 3

Apply the Kirchhoff's current law at the inverting terminal of the op-amp.

$$\frac{0-v_s}{R}+i_C=0$$

$$\frac{0-v_s}{R}+C\left[\frac{d}{dt}(0-v_O)\right]=0$$

Substitute the expression $L\frac{di}{dt}$ for output voltage v_O.

$$\frac{-v_s}{R}+C\left[\frac{d}{dt}\left(-L\frac{di}{dt}\right)\right]=0$$

$$\frac{-v_s}{R}-LC\left(\frac{d^2i}{dt^2}\right)=0$$

$$\frac{d^2i}{dt^2}=\frac{-v_s}{RLC}$$

Thus, the differential equation for $i(t)$ which is second order is $\boxed{\frac{-v_s}{RLC}}$.